Modern Academy
for Engineering and Technology in Maadi
Electrical Engineering

MODERN ACADEMY FOR ENGINEERING & TECHNOLOGY

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QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (ELCn425) Digital Signal Processing
Examiners: Dr. Samir M. Kamal & Dr. Sara M. Hassan
Branch: Communications & Computers
Time: 2 hours
Number of Pages: 1 Number of Questions: 5 Attempt all questions

Question 1 [8 Points] [LO's: 1]

a- In detail, what are the main characteristics of the continuous time sinusoidal signals? 3
b- For the following signal: $x_a(t) = A \cos(2\pi f_0 t)$, where, A=2 volt and f_0 =10 KHz.
(i) Check the periodicity of this signal and find its minimum sampling frequency. 2.5
(ii) Using the minimum sampling frequency obtained part (i), find x(n) for this signal and calculate the sample values for this signal, x(n), in one period. 2.5

Question 2 [8 Points] [LO's: 3]

a- Prove that the DFT, X(K), of a sequence x(n) is periodic. 1
b- For the following sequence: x(n) = {1, 1, 0, 1}:
(i) Using the FFT algorithm, find the frequency components of this sequence, X(k). 2
(ii) Check your results obtained in part (i) by calculating x(n) using the IDFT method. 2
(iii) Using your results, verify Parseval's theorem. 1.5
(iv) Calculate and plot the amplitude and the phase spectrums of X(k). 1.5

Question 3 [8 Points] [LO's: 3]

a- Describe the convolution operation, and deduce its compact relations. 2
b- For the following sequences: x(n) = {1, 1, 0, 1} and h(n) = {1, 0, 0, 1}
(i) Using the direct method, calculate their cross-correlation. 2
(ii) Find the o/p, y(n), of a system has an impulse response h(n) when applied to it i/p signal x(n). 2
(iii) Calculate the autocorrelation of the sequence x (n). 2

Question 4 [8 Points] [LO's: 4]

a- For LTIS define the following items: Stability, Causality and Linearity. 1.5
b- For the following transfer function, H(z), given by: $H(z) = \frac{z^2(z+1)}{(z-0.5)(z-1)^2}$
(i) Plot the pole - zero diagram of H (z). 1
(ii) Find the difference equation of this filter, y (n). 1.5
(iii) A digital system has the transfer function, H(z), given in part (b), using the partial fraction method by aiding of the given table-1, find the first "4" values for its impulse response, h(n). 4

Question 5 [8 Points] [LO's: 2, 4]

a- The coefficients of transversal filter are $h_0 = h_1 = -0.45$ and $h_2 = 0.75$. Draw the realization structure for the transversal filter. Find the impulse response, difference equation of the filter and draw its equivalent Lattice structure. 4 4
b- Check (✓) for the right statements and (X) for the wrong statements and correct the mistakes.
(i) FIR filters generally have first order poles. ( )
(ii) IIR requires more coefficients for sharp cut off filters than FIR. ( )
(iii) FIR filter output is a function in the present input, previous input and previous output ( )
(iv) FIR requires more coefficients for sharp cutoff filters than IIR. ( )

Table-1 Z-transforms table of some required sequences.

Discrete-time sequence x(n), n ≥0	z-transform X(z)	Note
kn	$\frac{kz}{(z-1)^2}$	k and α are constants
k	$\frac{kz}{z-1}$	
$k\alpha^n$	$\frac{kz}{z-\alpha}$	

Good Luck